

Computer Networks

Complex Computing Problem (CCP)

25th May 2024

Muhammad Alyan SP - 21319

Konain Raza SP - 21303

Aazeen Iftikhar SP - 21329

Shahzor Khan FL - 23851

Instructor: Mr. Hasan Shah

Contents

1.	Abstract.....	1
2.	LAN Setup Topologies	1
3.	VLAN Setup	3
4.	DHCP Configuration	5
5.	Web Server Configuration.....	6
6.	STP Configuration.....	7
7.	WAN Setup.....	8
8.	RIP Version 2	8
9.	DNS Setup	9
10.	PPP Encapsulation	10
11.	Testing.....	10

1. Abstract

This CCP report aims to depict the implementation of LANs (Local Area Networks) and WAN (Wide Area Network) in the Cisco Packet Tracer according to the requirements provided. General abstract of the documents the entailing points that have been covered in the report. These include topologies for LAN, DHCP configuration, Webserver configuration and STP setup for LAN. And as for WAN it includes implementation of RIP version 2 on all the routers in the network and DNS configuration.

2. LAN Setup Topologies

We have setup four LANs representing four different cities i.e., Rawalpindi, Islamabad, Lahore, and Karachi. Each LAN has a different topology in order to introduce diversity among the implementation of different concepts for each topology. Each LAN has further three VLANs (Virtual Local Area Networks) as per the requirements. Let's move on to how we setup the LANs and later portion will contain the information about how those LANs were connected in a WAN to establish communication among LANs of different cities. Figure 1, figure 2, figure 3 and figure 4 shows the topology for Rawalpindi LAN, Lahore LAN, Islamabad LAN, and Karachi LAN respectively.

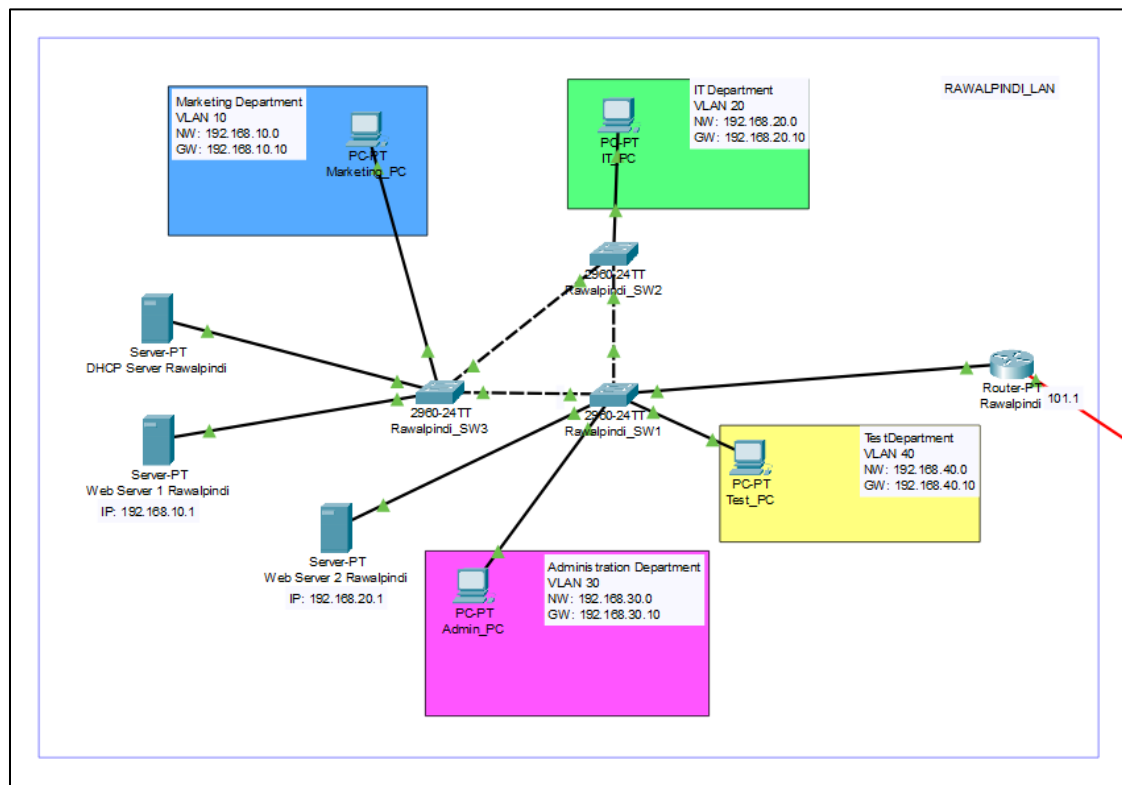


Figure 1 Rawalpindi LAN topology

Computer Networks (CCP)

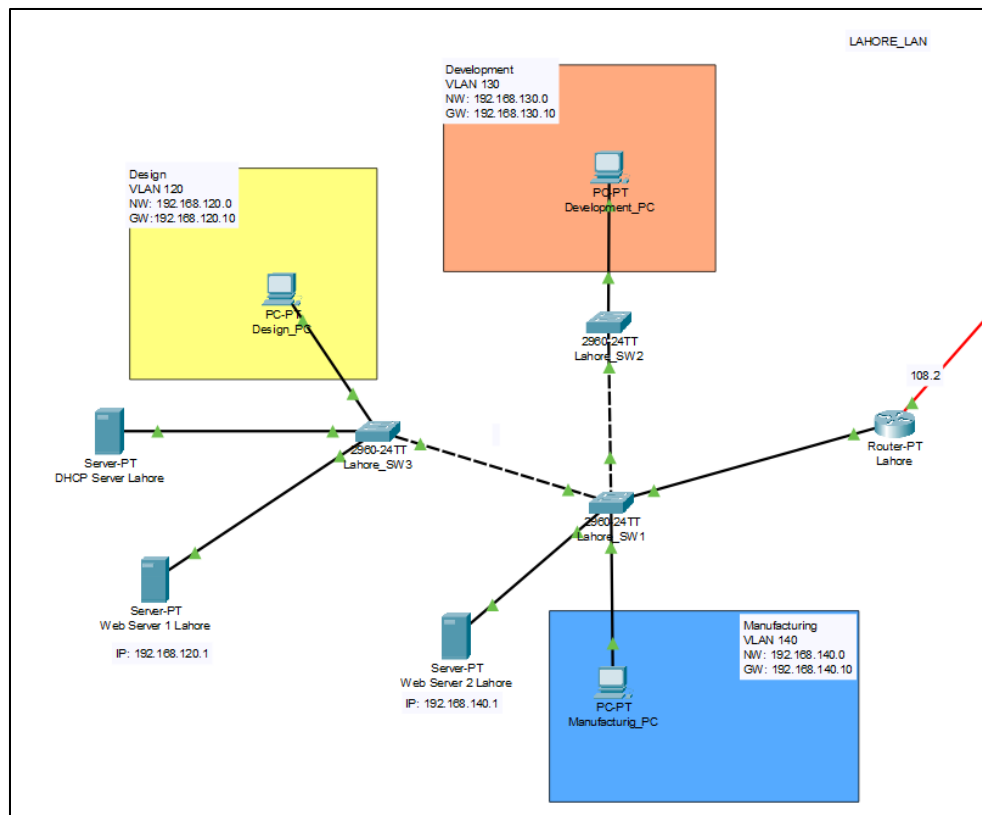


Figure 2 Lahore LAN topology

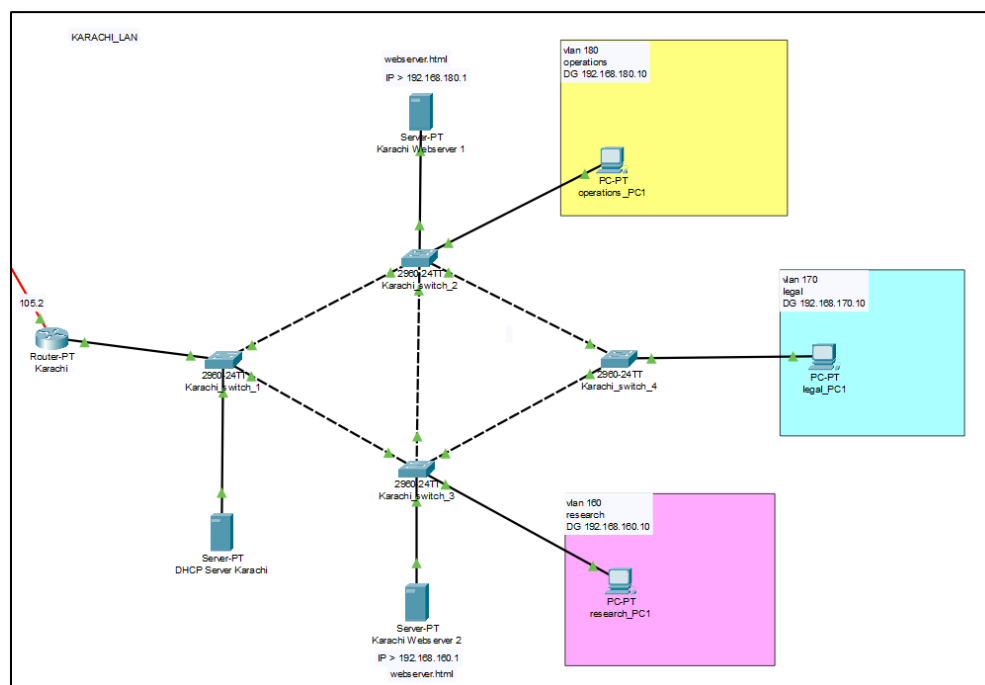


Figure 3 Karachi LAN topology

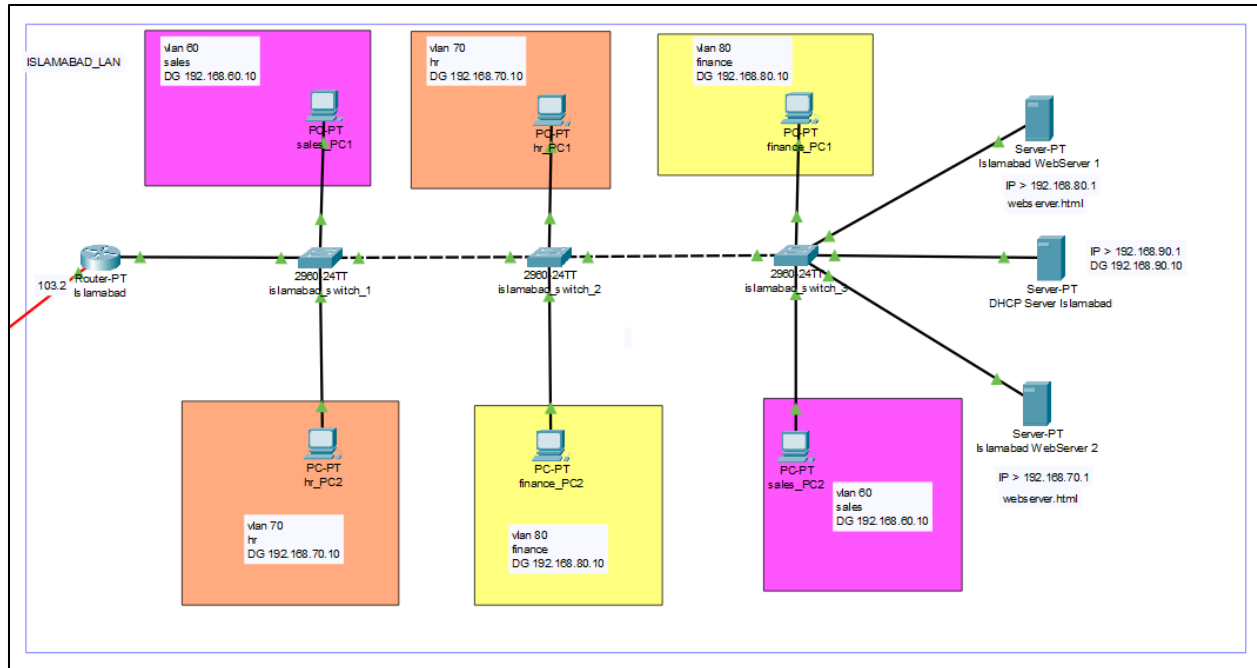


Figure 4 Islamabad LAN topology

3. VLAN Setup

After setting up the topologies, we needed to setup all the VLANs as per our requirements (minimum three LANs per LAN). We assigned the VLANs as follows for all the networks.

For Rawalpindi LAN;

1. VLAN 10 – Marketing department
2. VLAN 20 – IT department
3. VLAN 30 – Administration department
4. VLAN 40 – Test department
5. VLAN 50 – DHCP

For Islamabad LAN;

1. VLAN 60 – Sales department
2. VLAN 70 – HR department
3. VLAN 80 – Finance department
4. VLAN 90 – DHCP

For Lahore LAN;

1. VLAN 120 – Design department
2. VLAN 130 – Development department
3. VLAN 140 – Manufacturing department

4. VLAN 150 – DHCP

For Karachi LAN;

1. VLAN 160 – Research department
2. VLAN 170 – Legal department
3. VLAN 180 – Operations department
4. VLAN 190 – DHCP

Note: we have created one extra VLAN in all the LANs on which the DHCP server has been configured. Each LAN has three to four switches which have been named and numbered against the name of the city i.e., Islamabad_switch_1, Lahore_switch_2, etc. And the PCs have been named against their assigned VLAN i.e., marketing_PC1, legal_PC2, etc.

Steps in setting up VLANs

1. Creating VLANs on all the switches in a LAN.
2. Creating trunk mode between router and switch, and between switches.
3. Assigning interfaces of the switches to the VLANs.
4. Configuring router on a stick and assigning IP addresses to all the sub-interfaces created against each VLAN.

Following figures show the steps that were involved in configuring VLAN over Rawalpindi LAN.

```
Switch>
Switch>en
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname rawalpindi_switch_1
rawalpindi_switch_1(config)#
rawalpindi_switch_1(config)#vlan 10
rawalpindi_switch_1(config-vlan)#name marketing
rawalpindi_switch_1(config-vlan)#ex
rawalpindi_switch_1(config)#vlan 20
rawalpindi_switch_1(config-vlan)#name IT
rawalpindi_switch_1(config-vlan)#ex
rawalpindi_switch_1(config)#vlan 30
rawalpindi_switch_1(config-vlan)#name finance
rawalpindi_switch_1(config-vlan)#ex
rawalpindi_switch_1(config)#vlan 40
rawalpindi_switch_1(config-vlan)#name testing
rawalpindi_switch_1(config-vlan)#ex
rawalpindi_switch_1(config)#vlan 50
rawalpindi_switch_1(config-vlan)#name dhcp
rawalpindi_switch_1(config-vlan)#ex
rawalpindi_switch_1(config)#ex
rawalpindi_switch_1#
%SYS-5-CONFIG_I: Configured from console by console
rawalpindi_switch_1#
```

Figure 5 Setting VLANs on switch

After doing this, we configured router on a stick to establish inter-VLAN communication.

```
rawalpindi(config)#
rawalpindi(config)#int fa0/0.10
rawalpindi(config-subif)#encapsulation dot1q 10
rawalpindi(config-subif)#ip address 192.168.10.10 255.255.255.0
rawalpindi(config-subif)#ex
rawalpindi(config)#
rawalpindi(config)#int fa0/0.20
rawalpindi(config-subif)#encapsulation dot1q 20
rawalpindi(config-subif)#ip address 192.168.20.10 255.255.255.0
rawalpindi(config-subif)#ex
rawalpindi(config)#
rawalpindi(config)#int fa0/0.30
rawalpindi(config-subif)#encapsulation dot1q 30
rawalpindi(config-subif)#ip address 192.168.30.10 255.255.255.0
rawalpindi(config-subif)#ex
rawalpindi(config)#
rawalpindi(config)#int fa0/0.40
rawalpindi(config-subif)#encapsulation dot1q 40
rawalpindi(config-subif)#ip address 192.168.40.10 255.255.255.0
rawalpindi(config-subif)#ex
rawalpindi(config)#
rawalpindi(config)#int fa0/0.50
rawalpindi(config-subif)#encapsulation dot1q 50
rawalpindi(config-subif)#ip address 192.168.50.10 255.255.255.0
rawalpindi(config-subif)#ex
rawalpindi(config)#
```

Figure 6 Configuring Router-on-a-Stick

Once we were done with this, in order to establish inter-VLAN communication we had to assign IP addresses and Default gateways to all the PCs connected in the network. This could have been done two ways i.e., either by assigning static IPs or via DHCP server. We accomplished it via DHCP server.

4. DHCP Configuration

We attached a server and connected to one of the switches in the network. DHCP server was connected to the port of the switch that was assigned to one of the VLANs. For Rawalpindi DHCP server we connected it to VLAN 50. Then we assigned IP address and default gateway to the server as shown in the figure 7, and then we created server pools for all the VLANs as shown in the figure 8.

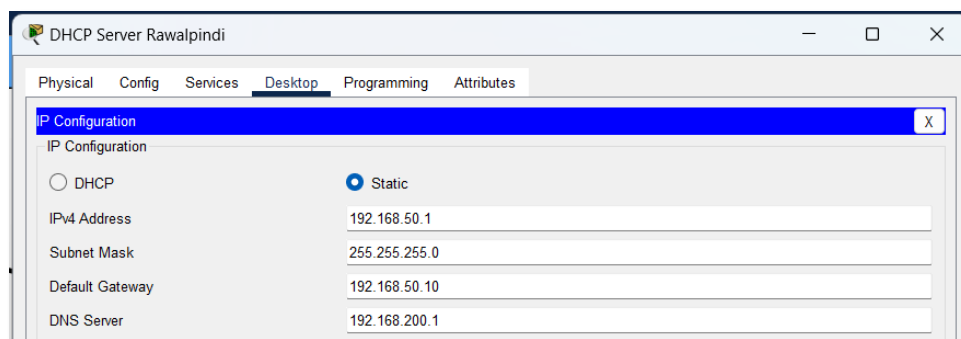


Figure 7 Assigning IP and Gateway to DHCP server

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
Pool20	192.168.2...	0.0.0.0	192.168.20.2	255.255.2...	254	0.0.0.0	0.0.0.0
Pool10	192.168.1...	0.0.0.0	192.168.10.2	255.255.2...	254	0.0.0.0	0.0.0.0
Pool50	192.168.5...	0.0.0.0	192.168.50.1	255.255.2...	255	0.0.0.0	0.0.0.0
Pool40	192.168.4...	0.0.0.0	192.168.40.1	255.255.2...	255	0.0.0.0	0.0.0.0
Pool30	192.168.3...	0.0.0.0	192.168.30.1	255.255.2...	255	0.0.0.0	0.0.0.0

Figure 8 Server Pools created over DHCP server

Once these steps of DHCP configuration were done and the intended PCs were connected to the ports that were assigned to their respective VLANs, we turned on DHCP service IP configuration for all the PCs, which automatically assigns to all the PCs over the network. This is shown in the figure 9 as follows.

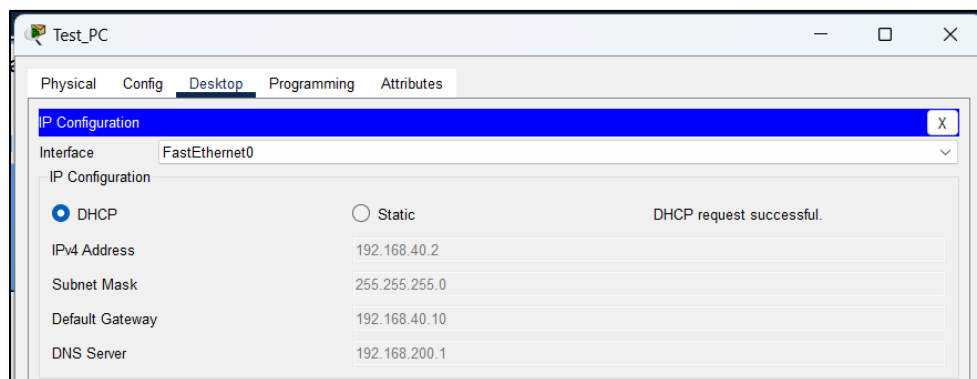


Figure 9 DHCP assigning IPs to the PC

After all these steps, inter-VLAN communication was successful.

5. Web Server Configuration

Next requirement that we needed to fulfill was to setup two web servers per LAN. We attached two servers and named them as Rawalpindi web server 1 and Rawalpindi web server 2. To configure web servers we needed to connect them to a particular VLAN like we did with the DHCP server, and assign them IP addresses and default gateways for that specific network. This could be done statically or via DHCP as we discussed above while configuring DHCP. But one problem that we faced during that configuration was that if DHCP assigned the IPs there was likely a chance for DHCP to assign a different IP to the server that would cause difficulty for us while we test the webserver configuration. So better option here for us was to assign the static IP and gateway for the webserver. After this we created one html page on each server as webserver.html these pages were successfully accessed from all the PC browsers in a network.

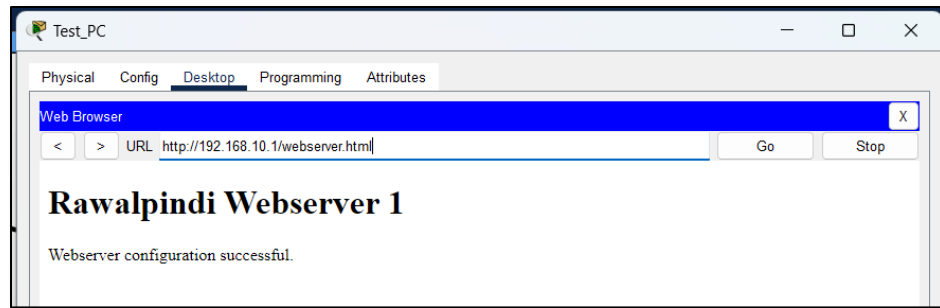


Figure 10 Accessing webpages from PC in LAN

6. STP Configuration

STP configuration was required where we were facing the looping problem in a network. As we designed different topologies to address this issue, STP configuration was not needed for all the networks. We configured STP for Rawalpindi LAN and Karachi LAN. Setup is shown in the figure 11 as follows. We select one switch as the root bridge, usually the switch with most connections is selected as a root bridge. After that we set the priority over the root bridge as shown in figure 11. We also select a secondary root bridge for which we set priority as shown in figure 12. (We did this for Karachi LAN, as it had a slightly complex topology than the Rawalpindi LAN). Figure 13 shows the priority setting for default bridges over the network.

```

rawalpindi_switch_1>
rawalpindi_switch_1>en
rawalpindi_switch_1#config t
Enter configuration commands, one per line. End with CNTL/Z.
rawalpindi_switch_1(config)#spanning-tree vlan 1 priority 4096
rawalpindi_switch_1(config)#end
rawalpindi_switch_1#
%SYS-5-CONFIG_I: Configured from console by console
  
```

Figure 11 Setting priority for the root bridge

```

karachi_switch_2(config)#
karachi_switch_2(config)#spanning-tree vlan 1 priority 8192
karachi_switch_2(config)#end
karachi_switch_2#
%SYS-5-CONFIG_I: Configured from console by console
  
```

Figure 12 setting priority for secondary bridge

```

karachi_switch_3(config)#
karachi_switch_3(config)#spanning-tree vlan 1 priority 12288
karachi_switch_3(config)#end
karachi_switch_3#
%SYS-5-CONFIG_I: Configured from console by console
  
```

Figure 13 Setting priority for default bridges

All the other LANs were setup in the similar manner as we showed for the Rawalpindi LAN. After we were done with configuring LANs started with WAN setup to establish inter-LAN communication.

7. WAN Setup

Once done with LANs we connected 5 routers in the following manner as shown in the figure 14. We named the routers in the WAN as cityname_wan_router i.e., Rawalpindi_wan_router, Karachi_wan_router, etc. We defined networks that we used to establish communication between the networks. These networks were assigned to all the routers using RIP (Routing Information Protocol) version 2.

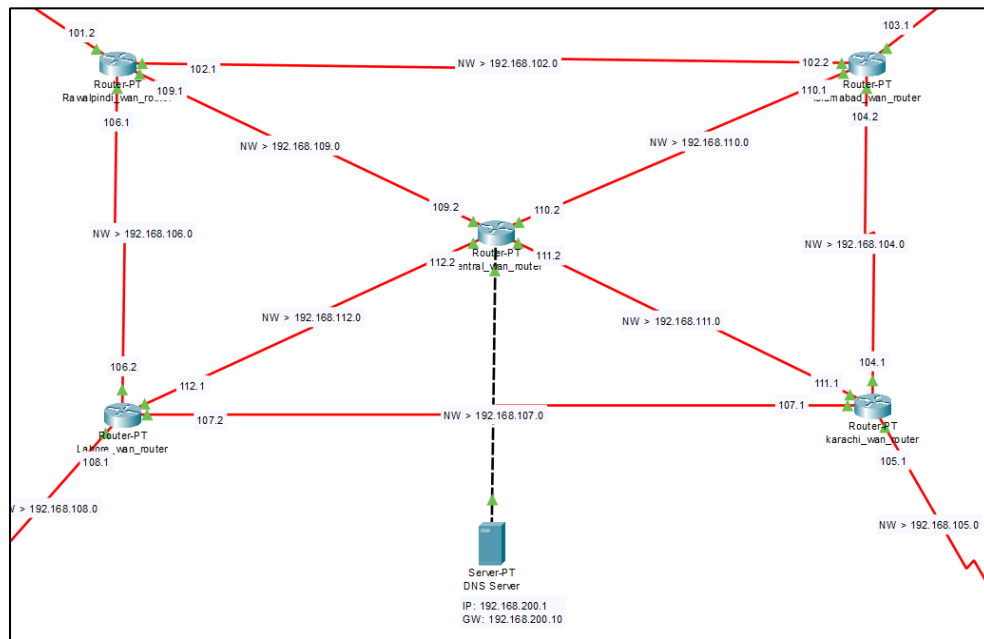


Figure 14 WAN topology

8. RIP Version 2

To establish communication we need to assign RIP version 2 over all the routers and also define the IP addresses for the ports that will be used for the communication. Figure 15 shows how we applied RIP version 2 to the routers.

```
lahore_wan_router(config)#router rip
lahore_wan_router(config-router)#ver 2
lahore_wan_router(config-router)#
lahore_wan_router(config-router)#network 192.168.106.0
lahore_wan_router(config-router)#network 192.168.107.0
lahore_wan_router(config-router)#network 192.168.108.0
lahore_wan_router(config-router)#network 192.168.112.0
lahore_wan_router(config-router)#
lahore_wan_router(config-router)#ex
lahore_wan_router(config)#
```

Figure 15 Implementing RIP version 2

Similarly, we defined RIP and networks for all the routers connected in the network.

9. DNS Setup

Moving on to the DNS requirement we connected a server to the central router. Configuration of DNS for the central router is comparatively easier as it is easily accessible by all the LANs. After connecting server to the router, we defined a network between DNS and the router and assigned IP address to the DNS against that network.

We also defined the network to the router as we did when we applied RIP version 2 to the router. This established the connection between DNS and central router (verified by the ping). After this we assigned the DNS IP address to all the DHCP servers in all the LANs so that could access the services of DNS. Upon refreshing the PCs, DHCP automatically defined the DNS IP address for all the VLAN PCs.

Then we entered hostnames against their IP addresses in the DNS server, so they could be accessed from all the LANs. And the last step was to test the working of DNS (shown in figure 16) and WAN (shown in figure 17).



Figure 16 Accessing Islamabad webserver from Lahore



Figure 17 Accessing Rawalpindi webserver from Karachi

10. PPP Encapsulation

Point-to-Point Protocol (PPP) is a widely-used data link layer protocol in networking, primarily employed to establish direct connections between two network nodes. PPP encapsulates network layer protocol information over point-to-point links, such as between routers in a Wide Area Network (WAN).

For our last requirement in WAN, we applied PPP encapsulation to all of the serial interfaces of the routers connected in WAN.

```
Central_wan_router>enable
Central_wan_router#
Central_wan_router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Central_wan_router(config)#interface Serial2/0
Central_wan_router(config-if)#
Central_wan_router(config-if)#exit
Central_wan_router(config)#interface Serial2/0
Central_wan_router(config-if)#encapsulation ppp
Central_wan_router(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to down

Central_wan_router(config-if)#no shutdown
Central_wan_router(config-if)#interface Serial3/0
Central_wan_router(config-if)#encapsulation ppp
Central_wan_router(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial3/0, changed state to up

Central_wan_router(config-if)#no shutdown
Central_wan_router(config-if)#interface Serial6/0
Central_wan_router(config-if)#encapsulation ppp
Central_wan_router(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial6/0, changed state to down

Central_wan_router(config-if)#no shutdown
Central_wan_router(config-if)#interface Serial7/0
Central_wan_router(config-if)#encapsulation ppp
Central_wan_router(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial7/0, changed state to up

Central_wan_router(config-if)#no shutdown
Central_wan_router(config-if)#
Central_wan_router(config-if)#ex
Central_wan_router(config)#ex
Central_wan_router#
%SYS-5-CONFIG_I: Configured from console by console

Central_wan_router#wr
Building configuration...
[OK]
```

Figure 18 Implementing PPP encapsulation

11. Testing

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	IT_PC	Design_PC	ICMP		0.000	N	0	(edit)	(delete)
	Successful	hr_PC1	legal_PC1	ICMP		0.000	N	1	(edit)	(delete)
	Successful	Rawal...	karachi_w...	ICMP		0.000	N	2	(edit)	(delete)

Figure 19 Successful Packet transmission between LANs